

The side indicator function above allows us to use a consistent expression for buy and sell orders. Many authors prefer to state the trading cost function separately for buys and sells as follows:

$$\text{Cost} = \begin{cases} \sum x_i P_0 - \sum x_i p_i & \text{Buys} \\ \sum x_i p_i - \sum x_i P_0 & \text{Sells} \end{cases}$$

In a later chapter (Chapter 8, Algorithmic Decision Making Framework) we expand on the order completion assumption and introduce ways investors can incorporate opportunity cost into the market impact model and decision making process. We show how investors can develop strategies to maximize the likelihood of executing an order within the desired price range (e.g., within their limit price) and hence minimize the probability of incurring opportunity cost due to adverse price movement.

The Almgren & Chriss model is comprised of three main components: temporary cost function, permanent cost function, and the market impact dissipation function. The temporary and permanent impact functions define how much the stock price will move based on the number of shares traded. The dissipation function defines how quickly the temporary price dislocation will converge or move back to its fair value (or in most situations, the new fair value which incorporates the permanent market impact cost).

Let us utilize a discrete time period random walk model. This process is described below.

Random Walk with Price Drift—Discrete Time Periods

Let the arrival price or starting price be P_0 .

The price in the first period is equal to the starting price plus price drift in the first period plus noise (price volatility). That is, $P_1 = P_0 + \Delta P_1 + \varepsilon_1$.

Here ΔP_j represents the natural price movement of the stock in the j^{th} period and is independent of the order (e.g., it would have occurred if the order was or was not transacted in the market), and ε_j is random noise (volatility) in the j^{th} period.

The price in the second period is,

$$P_2 = P_1 + \Delta P_2 + \varepsilon_2$$